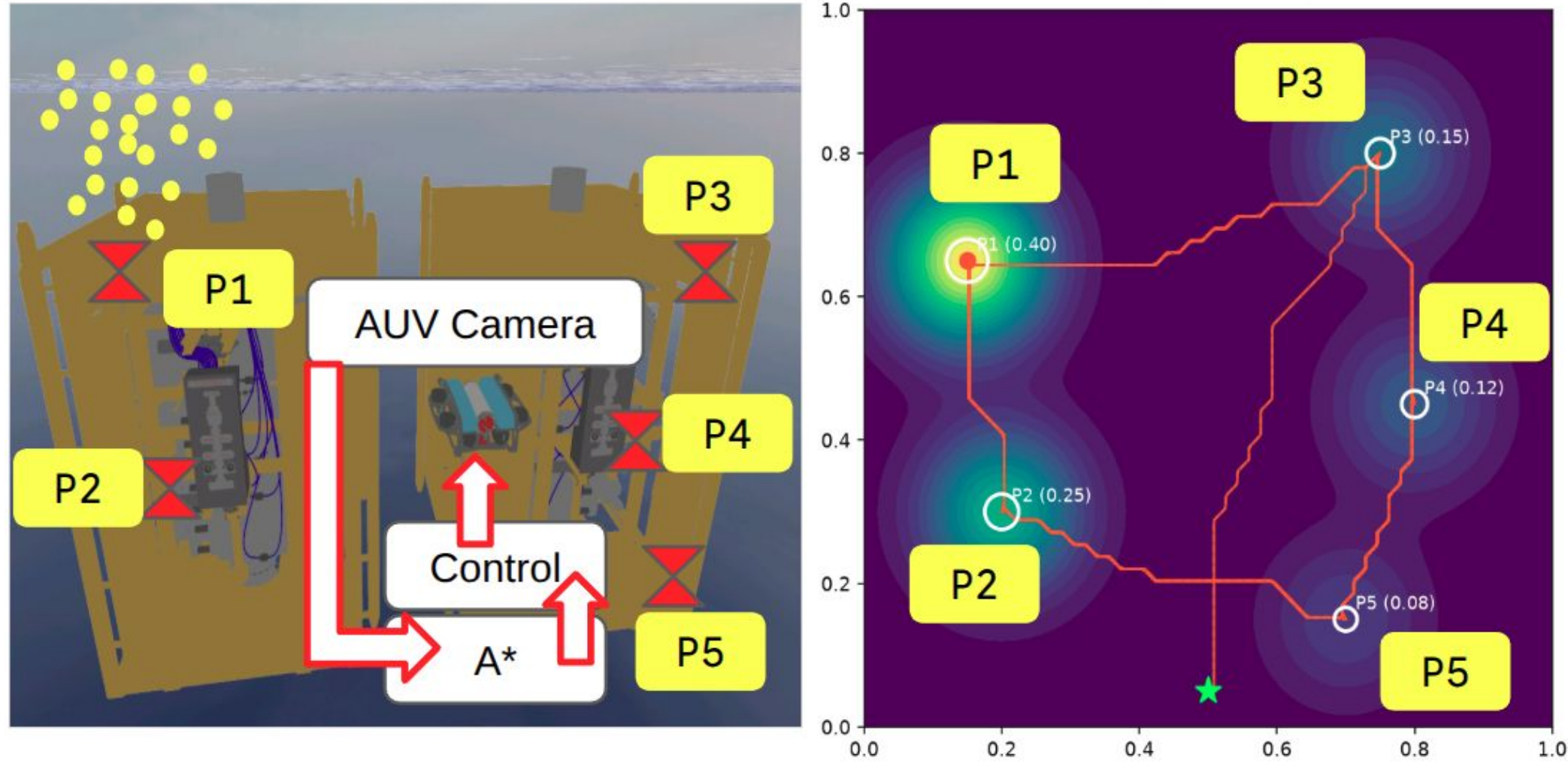
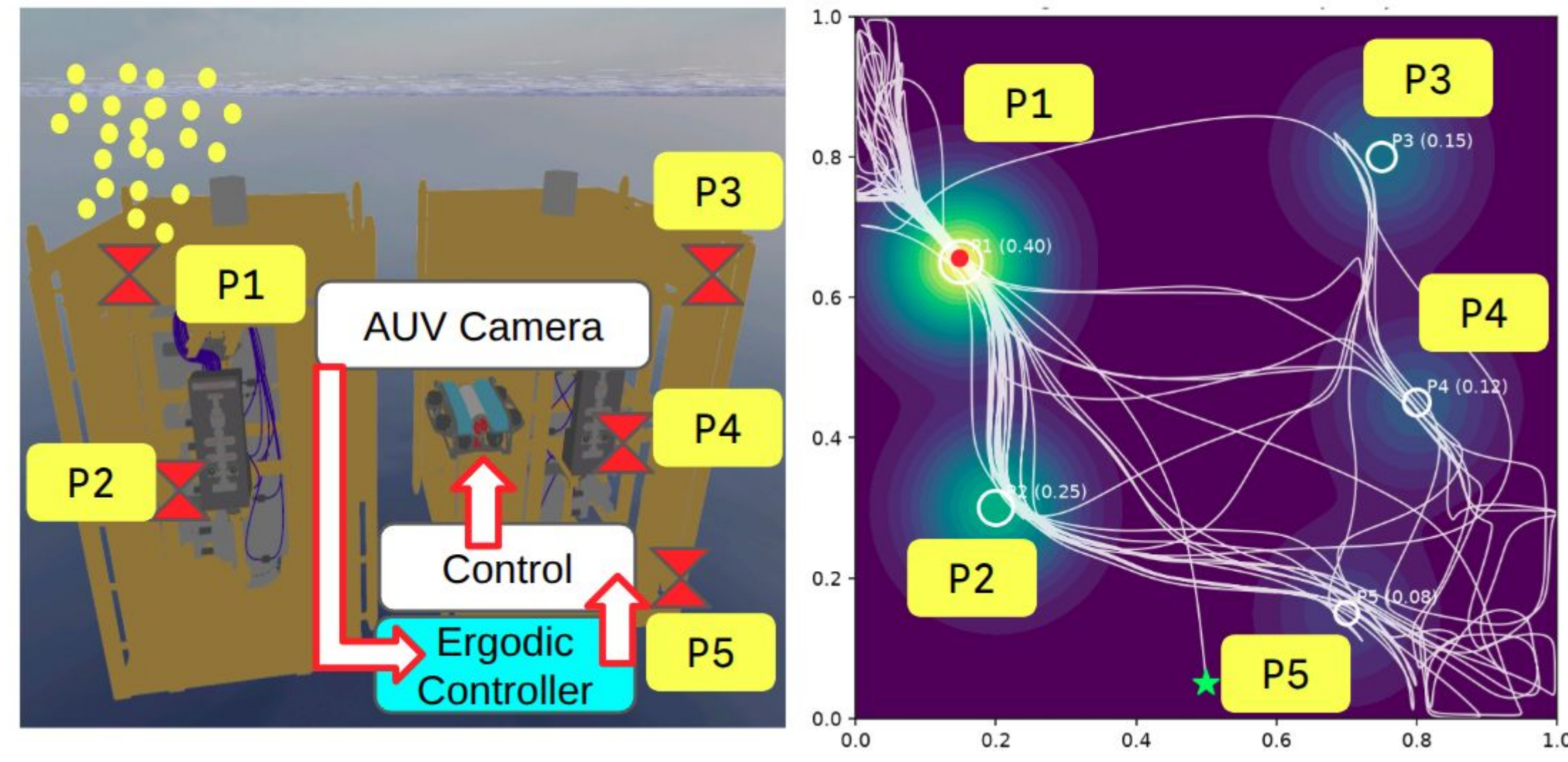


MOTIVATION

Baseline: Planned-Route Loop



SAGE (Ours): Autonomous Coverage Loop



Subsea infrastructure inspection often follows fixed routes that ignore changing risk conditions.

SAGE (Semantic and Adaptive Generative Ergodicity) replaces static inspections with adaptive coverage control, dynamically prioritizing inspection time and revisits based on live leak detection.

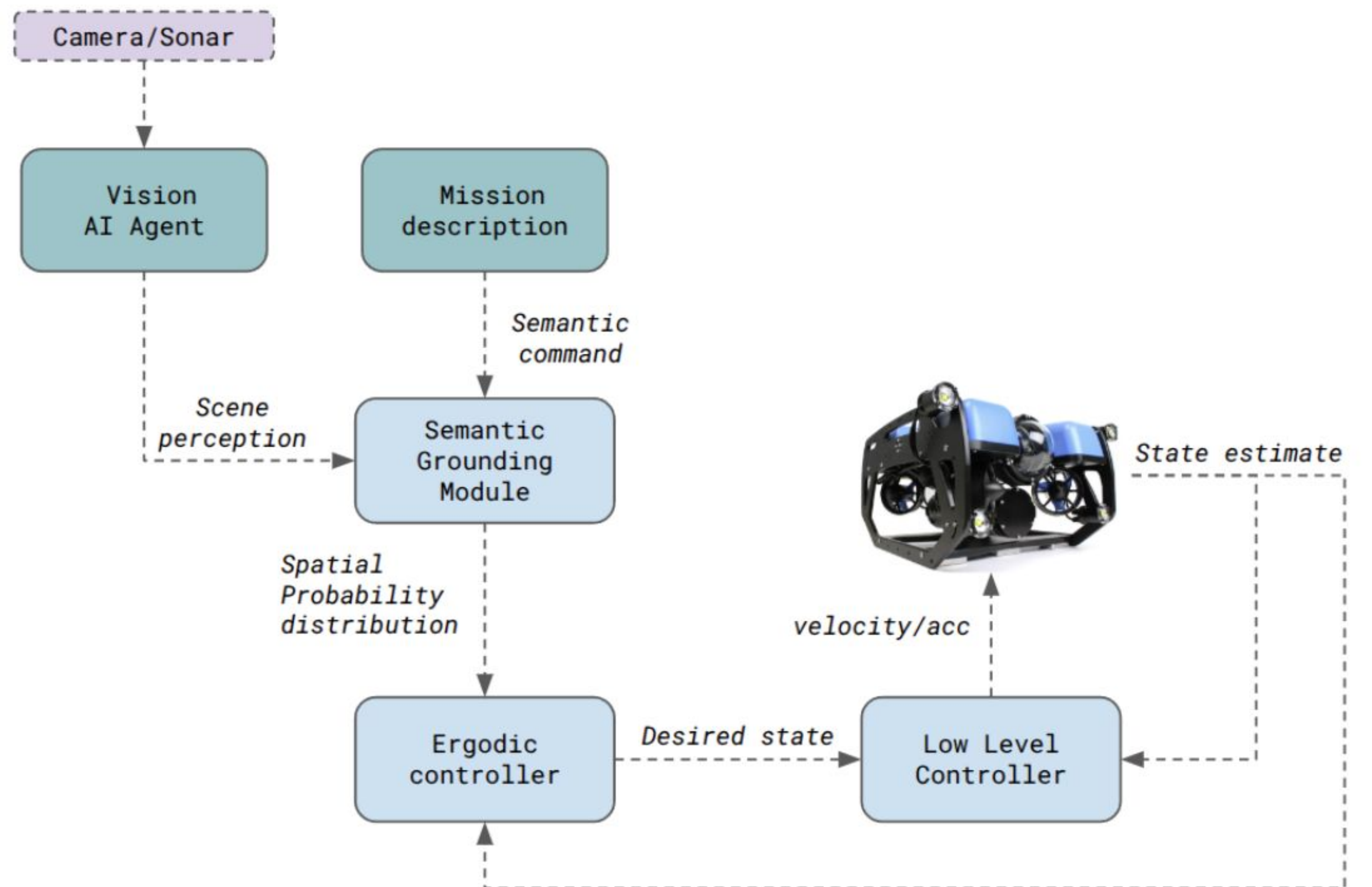
THE MAIN CONTRIBUTION

Vision AI Agent: Detects defects and leaks from camera and sonar data.

Semantic Grounding: Builds a live spatial probability map from mission goals and sensor data.

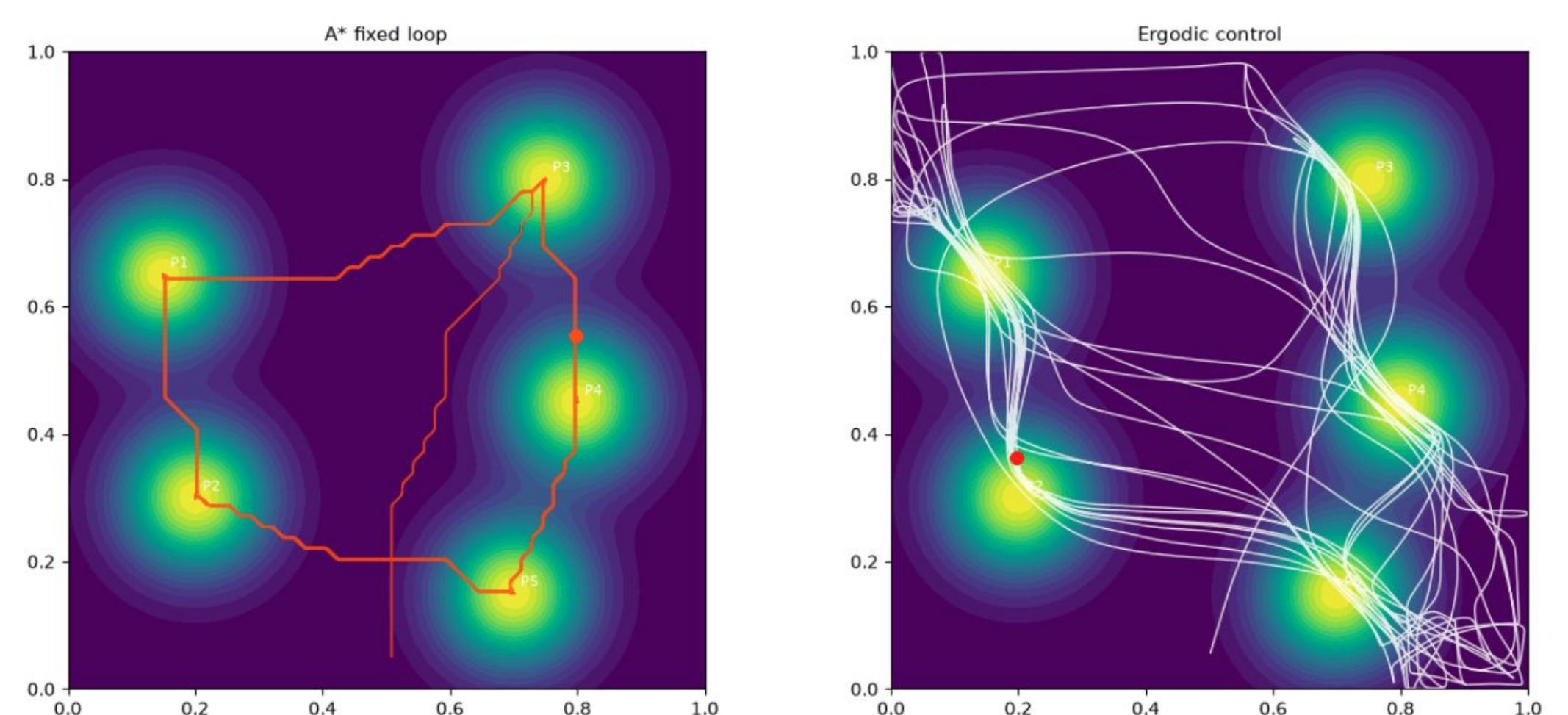
Ergodic Controller: Adapts coverage in real time without route re-planning.

Low-Level Controller: Executes command using continuous state feedback.

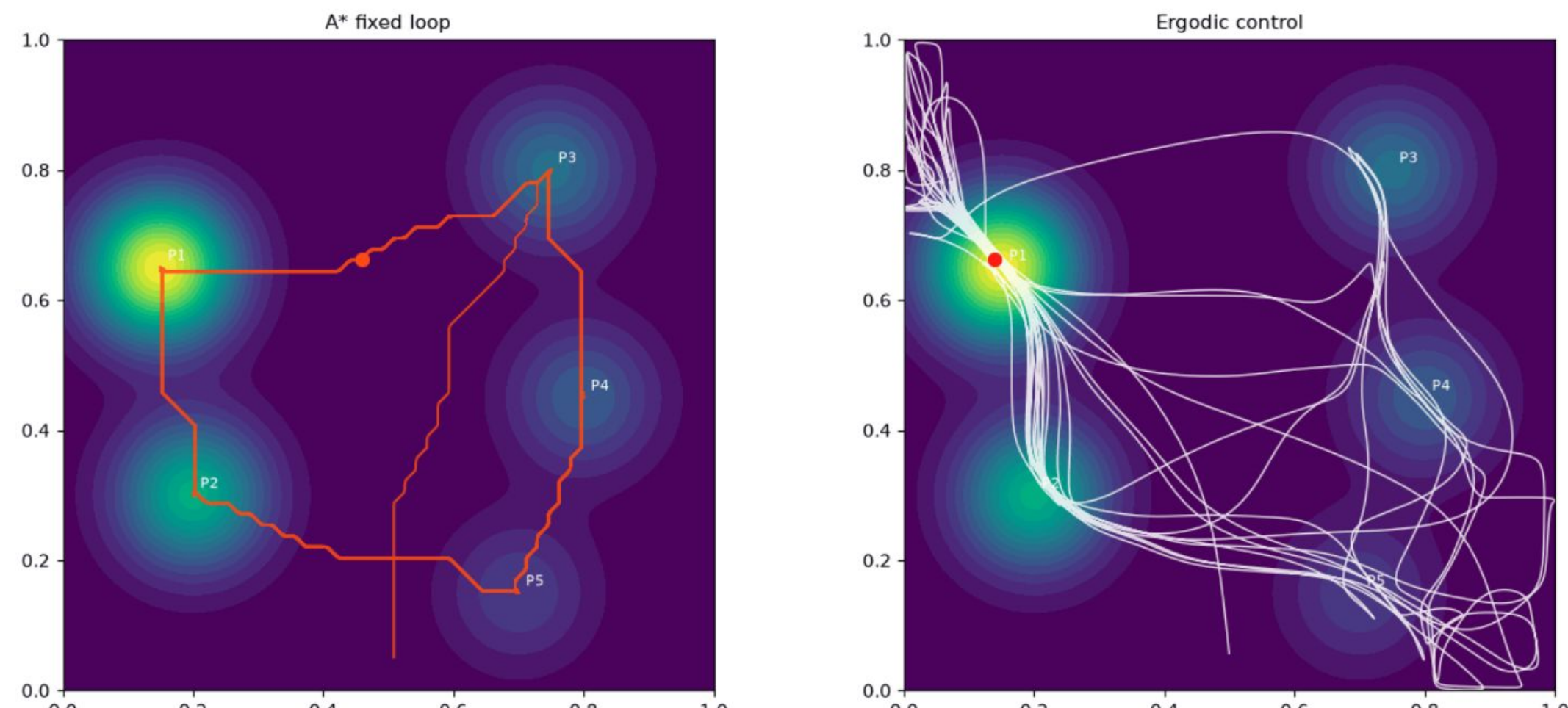


KEY RESULTS

Equal Weights (Scenario 1): Both methods cover all 5 valves evenly.



Graded Weights (Scenario 2): Inspection automatically focuses more on higher-risk valves (+0.88 correlation).



Leaking Valve (Scenario 3):
1.4x Faster Revisit: 5.8s vs. 8.1s with A*. More Coverage: 50 visits vs. 37 with A*. Instant Adaptation: Re-routes automatically in real time.

